THE EVERYDAY CITY

PHOTOVOLTAIC AGRICULTURE / COMPACT LIVING / PUBLIC LIFE

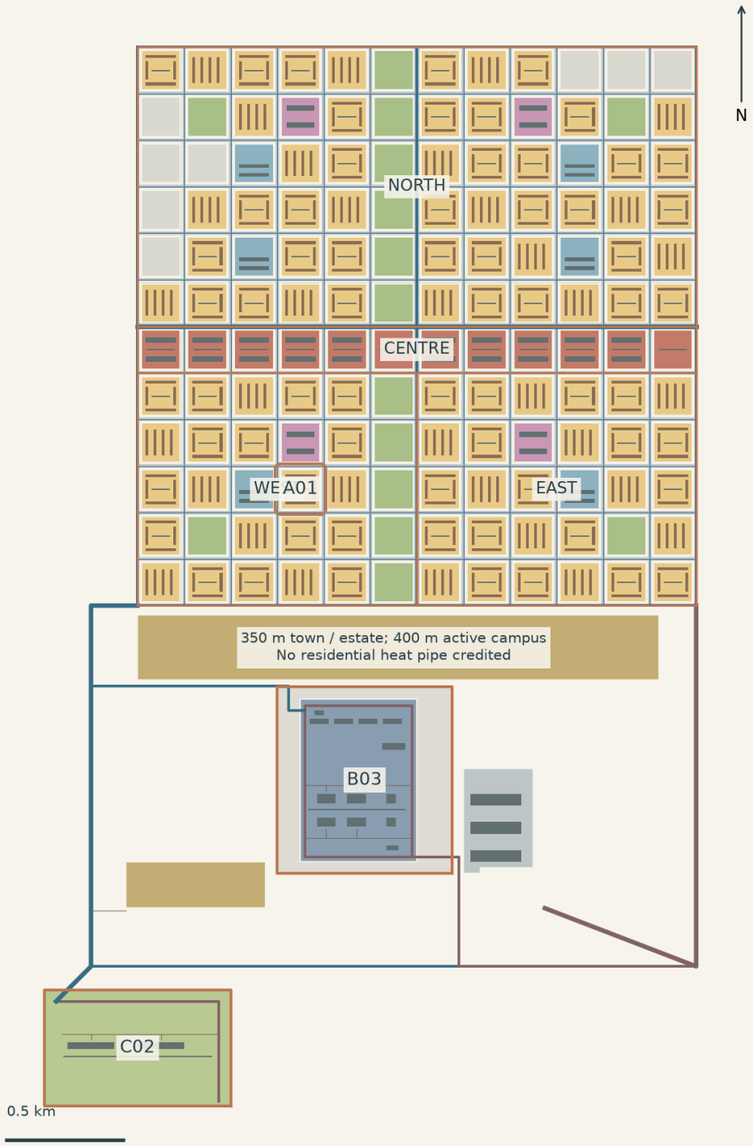


C02 / CONNECTED FARM-WORKSHOP EDGE / CLOSE PARTIAL AERIAL

A continuous town of homes, schools and daily services. Production supports the city from its own campus. The cover shows the proposed farm-learning edge; the overall plan locates it within the productive landscape and town.

A productive territory with everyday public life

V5 corrects territorial function before composing local images. The town, production campuses and farm interface retain their actual coordinates.



01 / Photovoltaic agriculture leads the territory

V5 covers the whole issued boundary. Agriculture and raised solar can share ground; the shared-field overlay is distinct from exclusive land and installed work. Mountain and drainage areas remain visible.

02 / Right-size industry and gateways

The former 252 km2 industrial and 292 km2 logistics regions contract to a48 km2 processing corridor and30 km2 of compact gateway reserves. No new jobs or cargo capacity is invented from their area.

03 / Compact living with independent services

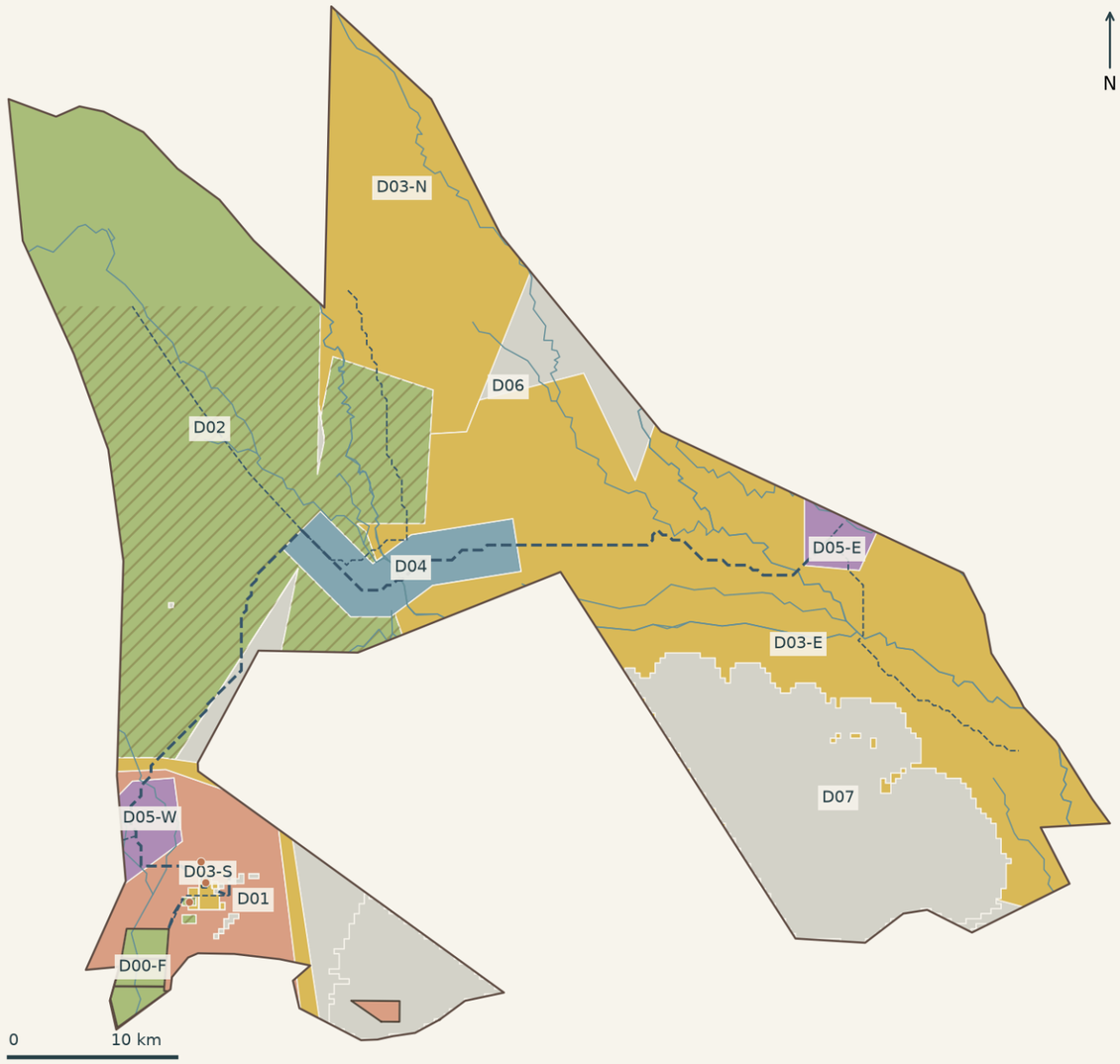
The complete2.4 x2.4km town is a detailed urban enlargement within the larger living territory. Its local housing, schools, civic band and separate compute campus remain staged operating scenarios.

A productive territory with compact living

V5 / exclusive base land on the issued boundary. Fixed pilots and retained mountains/drainage remain explicit; hatching marks conditional solar/crop co-use.

A PRODUCTIVE TERRITORY, WITH COMPACT LIVING

Whole issued site / V5 exclusive base land + co-use overlay / all allocations are resource-gated



ENERGY + COMPUTE
636.7 km² / 38.4% exclusive

AGRICULTURE
515.5 km² / 31.1% exclusive

LIVING + SERVICES
81.0 km² / 4.9% exclusive

INDUSTRY
48.2 km² / 2.9% exclusive

TRADE + GATEWAYS
29.6 km² / 1.8% exclusive

TERRAIN + DRAINAGE
347.0 km² / 20.9% exclusive

HATCH / solar + crop co-use candidate
362.2 km² new + 13.9 km² fixed context
Do not add it to base land.

Dashed: proposed regional road studies
Blue: source seasonal channels
Dark outlines: unchanged fixed pilots
A01 / B03 / C02: southwest detail views

D01 living | D02 agriculture | D03-S/N/E energy | D04 processing corridor | D05-W/E compact gateways | D06/07 retained landscape

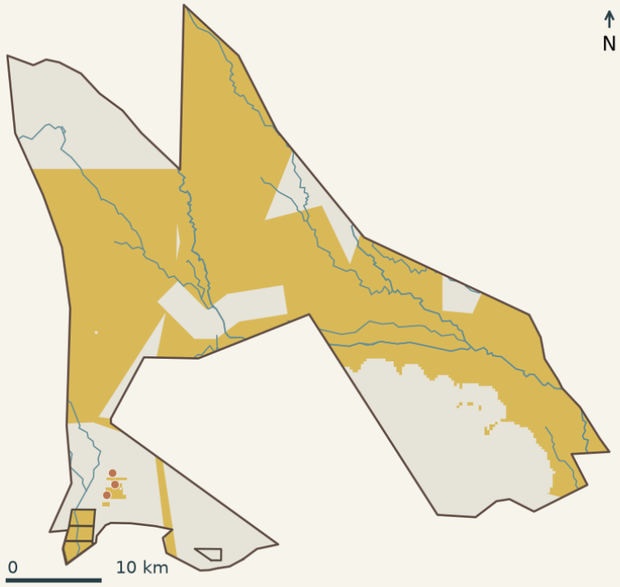
Respond to the whitepaper; distinguish the accounts

Reference ~60/30/5/3/2 is strategic. V5 compares functional coverage separately from exclusive land and current works.

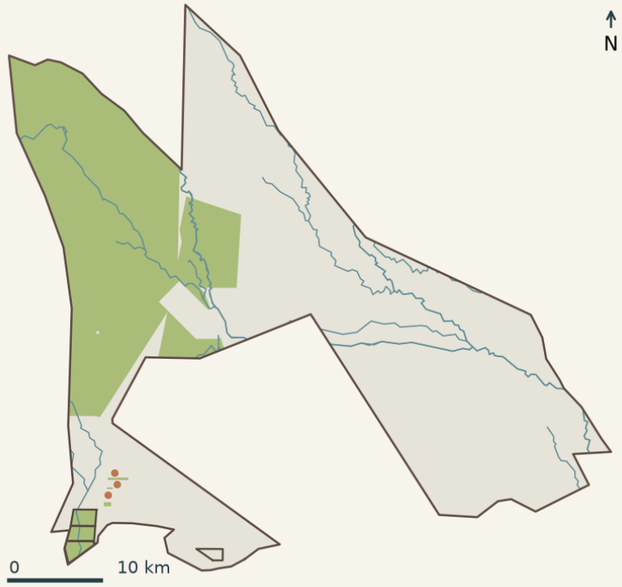
FIVE FUNCTIONS / THREE DIFFERENT ACCOUNTS

Whitepaper reference is strategic; exclusive land and overlapping coverage must not be confused.

ENERGY COVERAGE / 61.1%



AGRICULTURE / 31.1%



RETAINED / 20.9%



ENERGY + COMPUTE	AGRICULTURE	LIVING	INDUSTRY	TRADE / PORT
REFERENCE ~60%	REFERENCE ~30%	REFERENCE ~5%	REFERENCE ~3%	REFERENCE ~2%
V5 61.1%	**V5 31.1%**	**V5 4.9%**	**V5 2.9%**	**V5 1.8%**

The 376.1 km² solar/agriculture intersection is counted in both functional systems, once in base land. Coverage does not sum to 100%.

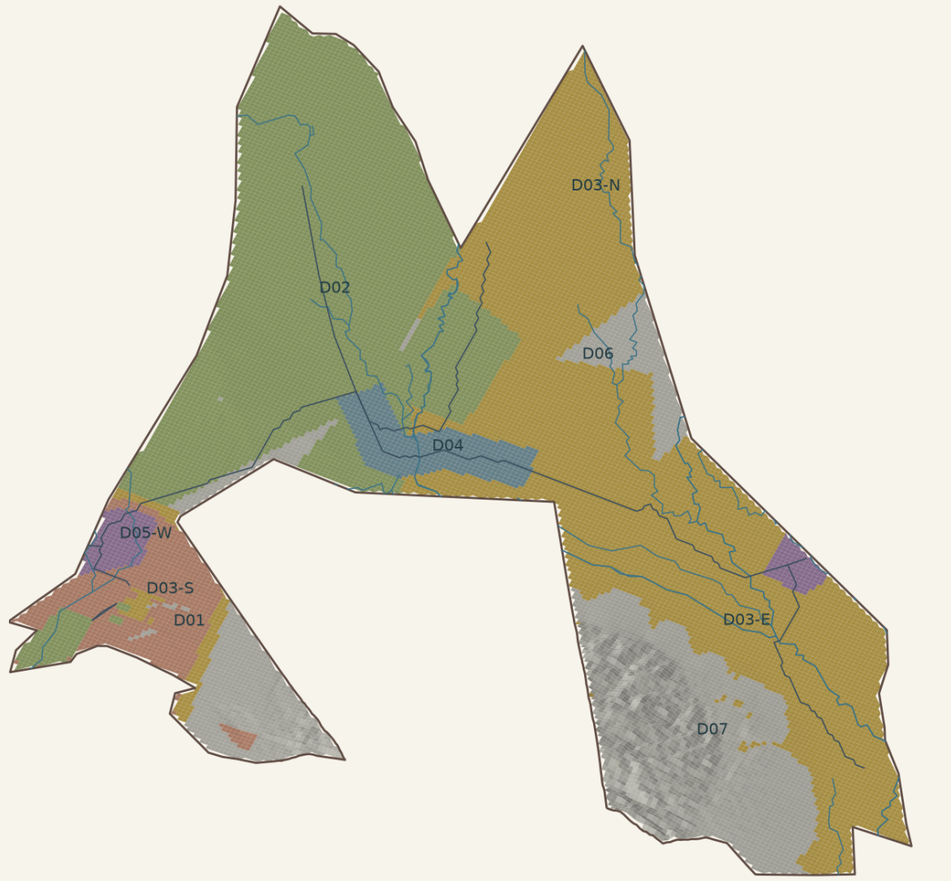
CURRENT WORKS / 7.6368 km² study footprint; 7.656 ha of new crop bands. Territorial envelopes do not add homes, IT MW or irrigation rights. Retained mountains and drainage remain visible. Future co-use requires soil, water, grid and access verification before a field is commissioned.

Terrain, source channels and allocated territories

Formal source-derived axon / whole issued boundary / true relative terrain scale. Gross district roles do not mean installed equipment or permission to build.

WHOLE-SITE TERRAIN + FUNCTIONAL STRUCTURE

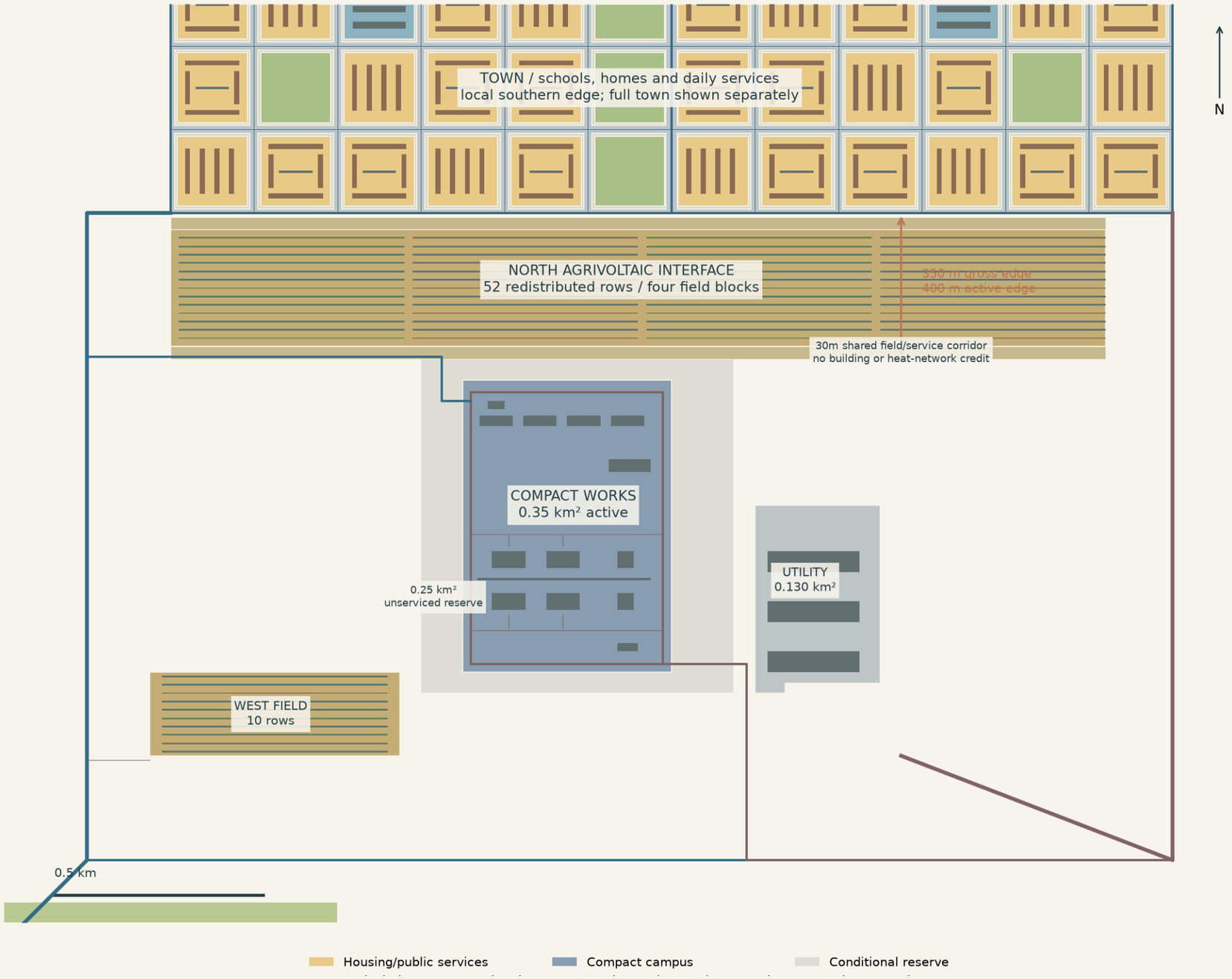
Source terrain in true relative scale / issued boundary / source seasonal channels / proposed route studies



Colours show allocated roles, not installed fields. Fixed pilots are retained. Local town/compute/farm rendering cameras remain local views.

Replace the missing middle with an operating interface

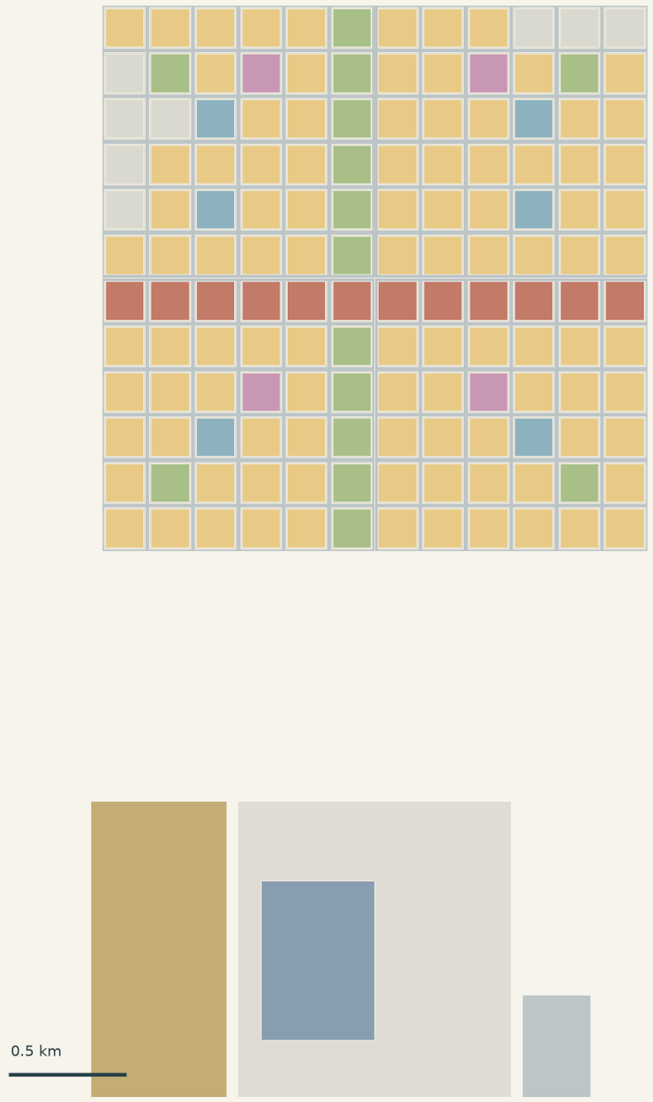
V5 / same housing, jobs, IT load and crop-net area. Move production1,050m north, shrink the estate, redistribute existing rows; do not fill the picture with invented capacity.



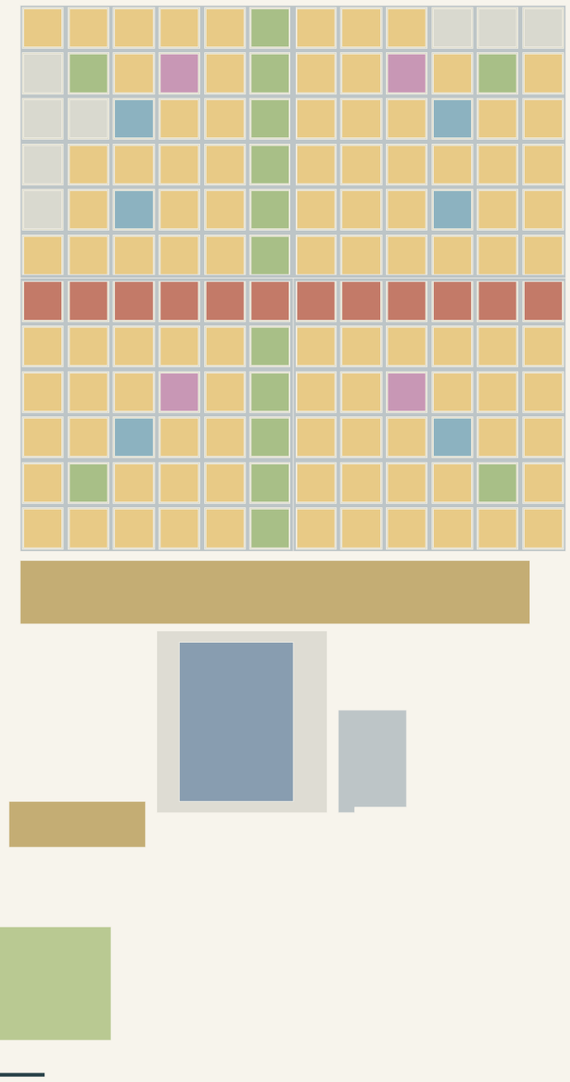
The land arrangement changed, not just the crop

Same scale before and after / current layout trial; noise, heat, fire and utility constraints remain to be checked.

BEFORE / 1.1 km unresolved estate gap



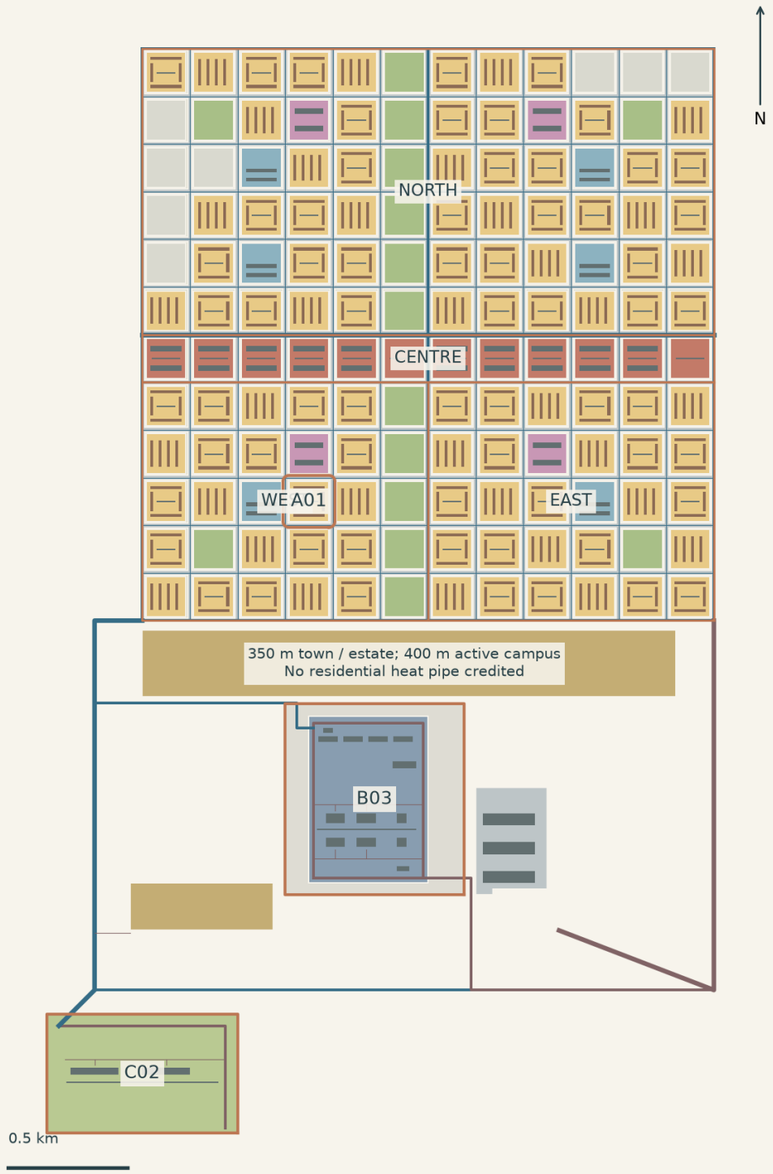
V5 / productive interface + compact campus



Same housing, jobs, IT load,62 main solar rows and crop-net area. Production/farm move1,050m north. Gross works estate1.56 →0.60km².
Neither the old1.1km nor new350m gross separation is an engineered noise/fire/thermal buffer; specialist review is required.

Choose a coherent town, then accept its distances

The earlier compact all-functions rectangle is withdrawn. The new scheme keeps the southwest access basin while giving housing a complete and independently serviced urban footprint.



The selected urban ground

Town: 5.76 km2; median screened slope 1.18%, 95th percentile 4.25%. Three housing cells touching the preliminary flood screen are held without capacity.

The compact interface replaces the unresolved gap

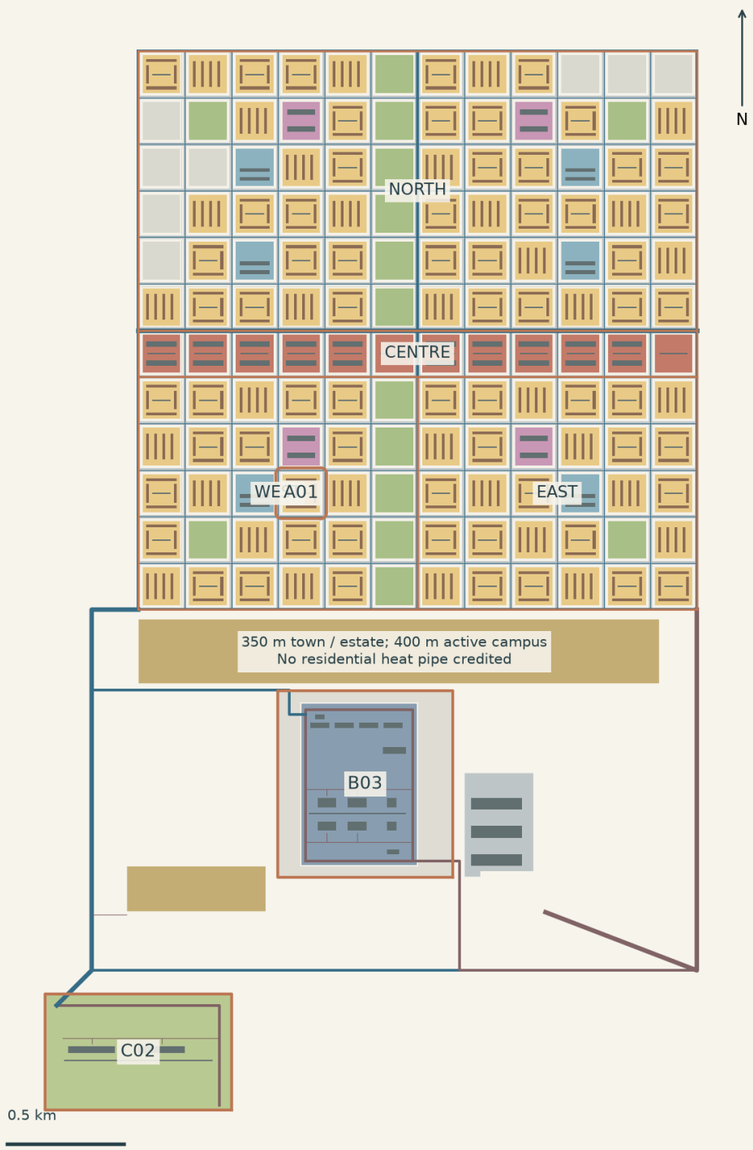
A compact co-location could shorten utilities and staff trips. The350m gross edge and400m active-campus distance contain the north agrivoltaic interface, service corridor and dry town verge. This is a compact layout trial; noise, heat and fire review can change it.

What could reverse this decision

Geotechnical conditions, a lawful access corridor, measured noise/dust, utility connection costs or actual demand may change the town or campus boundary. The 150 m channel test is a declared screening distance, not a statutory setback.

One settlement system, several operating territories

City scale / true relative coordinates / all phases. The study footprint totals 7.6368 km2; it is distinct from the full competition territory.



TOWN / residents and public life

Homes, shops, care and schools are organised by living district. The civic band is a shared destination; the north-south dry commons is public ground, not a lake.

WORKS / production and utility services

Four industrial data-centre halls at2 MW operating IT each are released 2 + 1 + 1. Repair, metered local drying trials and utilities have their own servicing. The adjacent 26.784 MWp PV scenario is conditional on module layout and grid study.

FARM / public learning and testing

Two farm-service sheds and a public shade promenade face eight new trial rows. Their small trial capacity is excluded from the main city PV account; crop bands have an explicit assumed water budget; commercial yields remain uncredited.

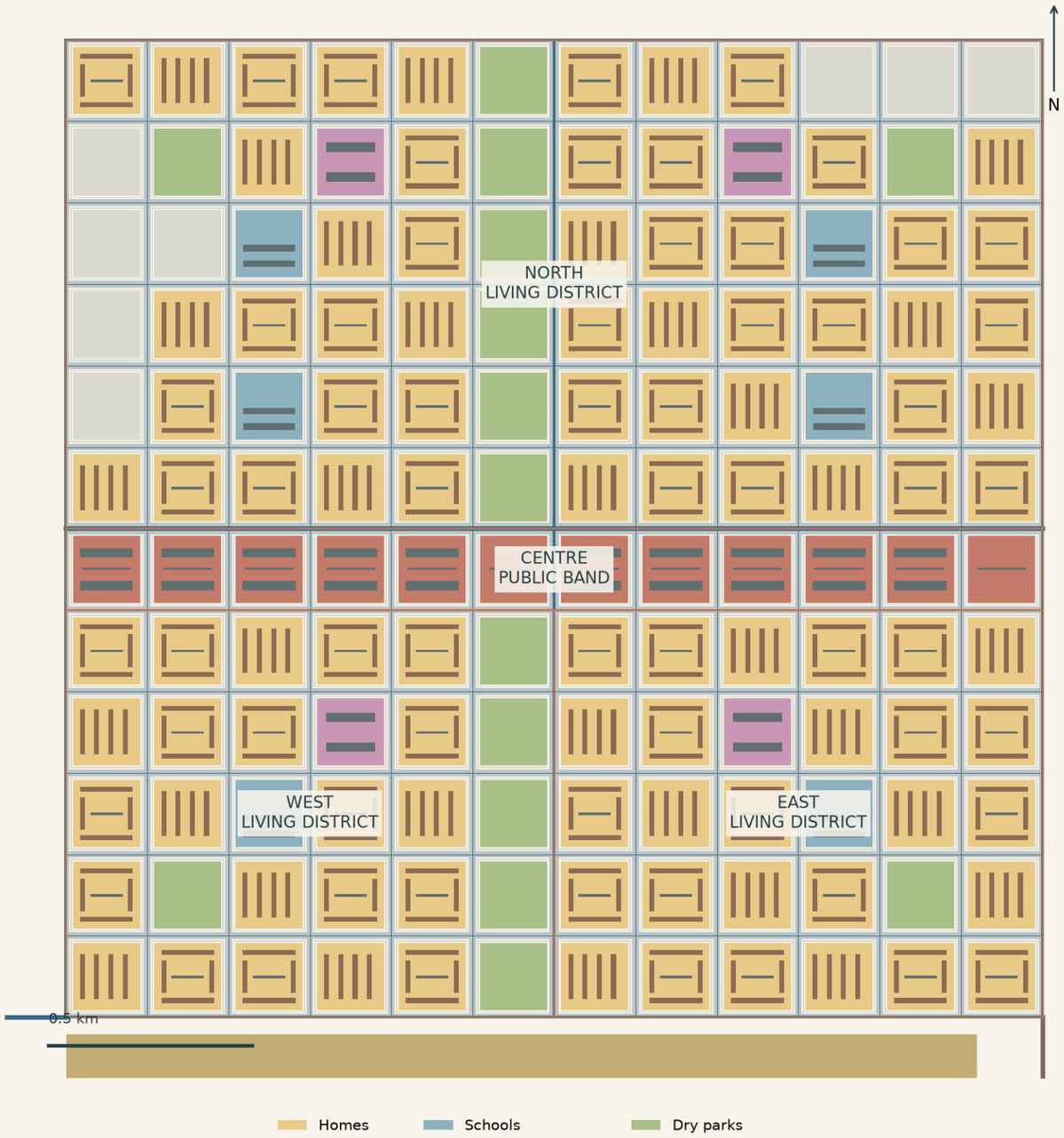
The town and its productive edges

V5 compact southwestern town edge and production interface / partial local frame / full-build-out scenario. Newly registered camera; not the entire territory or every outlying facility.



The north agrivoltaic band and compact campus meet the town edge at their actual coordinates. The C02 farm-learning node lies outside this partial frame; its location and views remain separate.

Districts, blocks and the civic band

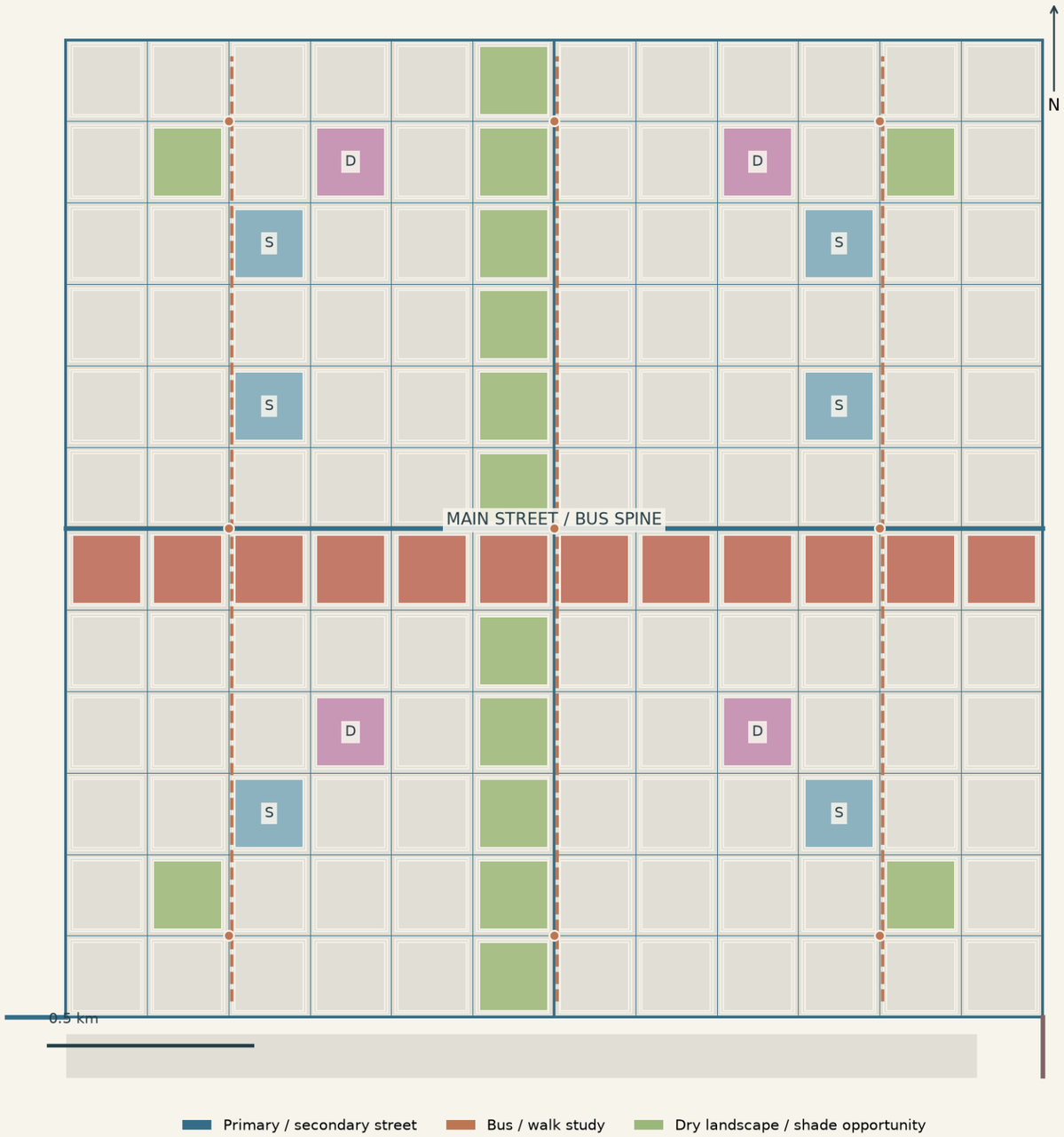


Three living districts.
The West is the first complete occupation area; the East extends the same service logic; the North completes the town.

Two public axes.
The east-west civic band holds shared destinations. The north-south dry commons gives a continuous public address.

Different housing grain.
Three-storey open court bars alternate with two-storey terrace lanes. The all-phase inventory yields 7,872 dwellings, a capacity ceiling rather than a proven market.

A hierarchy of streets and public movement

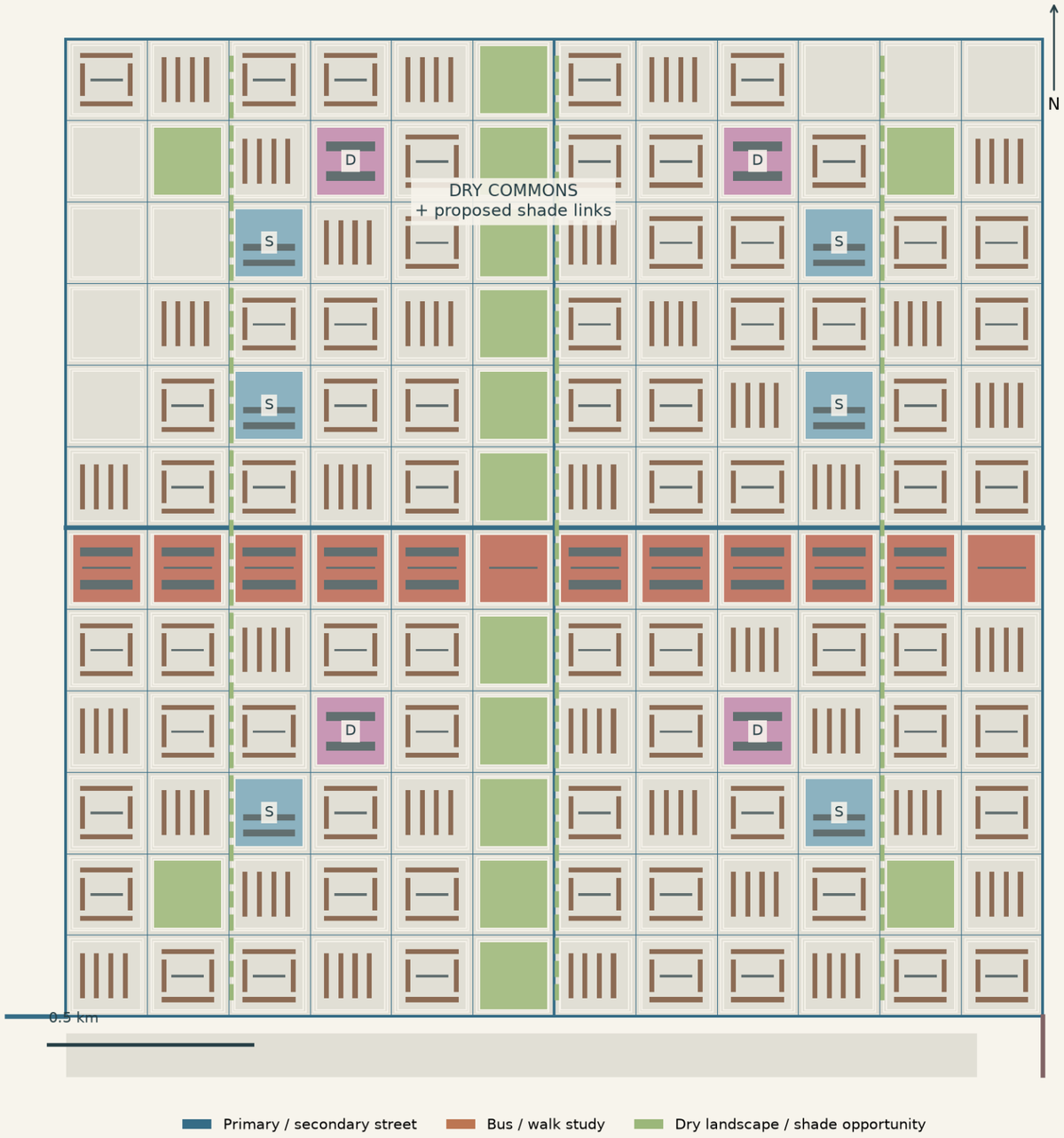


Blue: streets.
The 32 m civic Main Street and 28 m north-south avenue organise 20 m local streets. Widths are design reservations, pending engineering.

Rust: bus/walking study.
Stops mark a proposed network; frequency, accessibility and funding are not yet secured.

Outside the town.
A public southwest approach serves the farm and worker bus link. Production freight approaches on the east. Final junctions and emergency routes require transport review.

Dry commons and a shade network

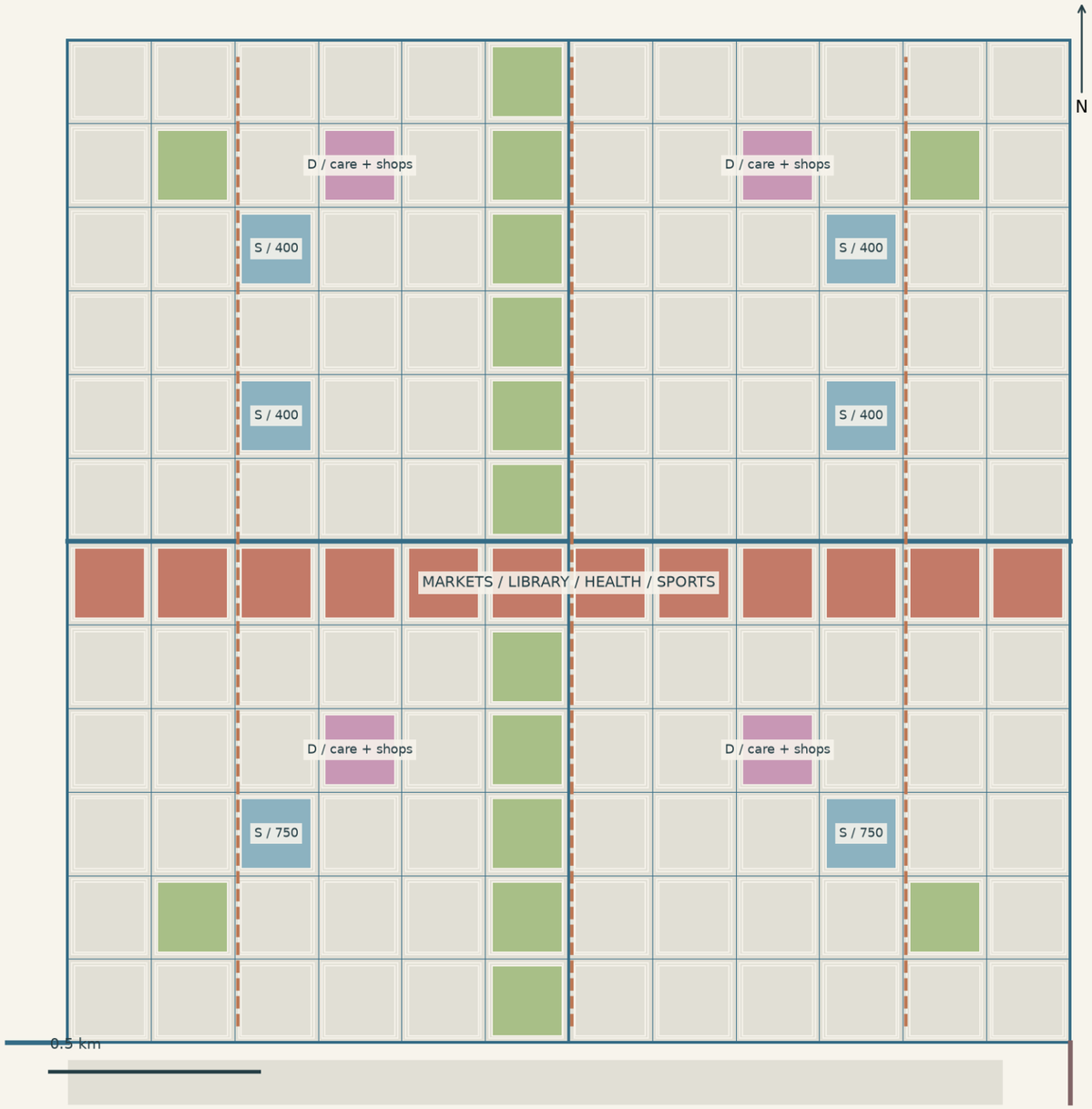


Landscape has different roles.
Dry parks and the commons provide public ground.
Source seasonal channels remain landscape/conveyance studies at territorial scale.

Green dashed links are proposals.
They identify future walking/shade opportunities on real street alignments, not established trees or guaranteed continuous shade.

Engineered shade first.
Courtyard, civic, worker and learning canopies are modelled explicitly. No irrigated lawn, decorative lake or mature planting capacity is credited.

Schools, care and daily destinations



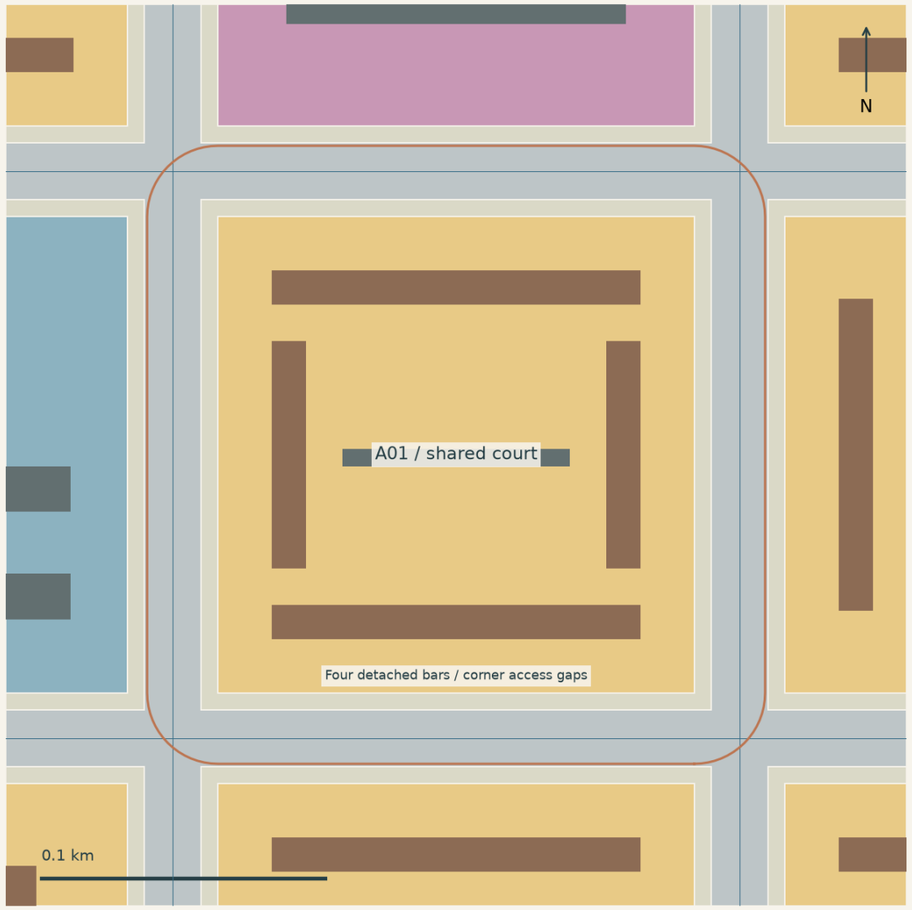
Services precede homes.
Two southern schools reserve 750 places each. Four northern schools reserve 400 each. The 3,100-place scenario exceeds an assumed 15% of 19,680 residents (2,952); demand, staffing and age mix need verification.

Four daily centres.
Care and neighbourhood shops supplement the larger markets, library, sports and health uses in the civic band.

No blanket five-minute claim.
Some trips across the 2.4 km town need the proposed bus and a measured safe-route audit.

A block is shared ground, not an employer compound

Two original housing types / build-out capacity from modelled buildings / detailed dwelling accessibility and tenure design remain next work.

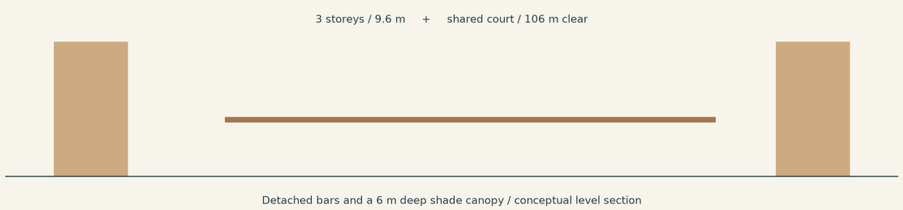


OPEN COURT BARS

Four separate bars define an open court. The two long bars each carry 30 dwellings; the two short bars each carry 18: 96 homes per court-block at three storeys. Corner gaps preserve public access; a 6 m deep canopy offers a shared address.

TERRACE LANES

Four two-storey rows each carry 12 dwellings: 48 homes per block. They offer a lower-rise counterpart, not a translated copy of the courtyard. Street frontage, emergency access and maintenance remain shared municipal obligations.

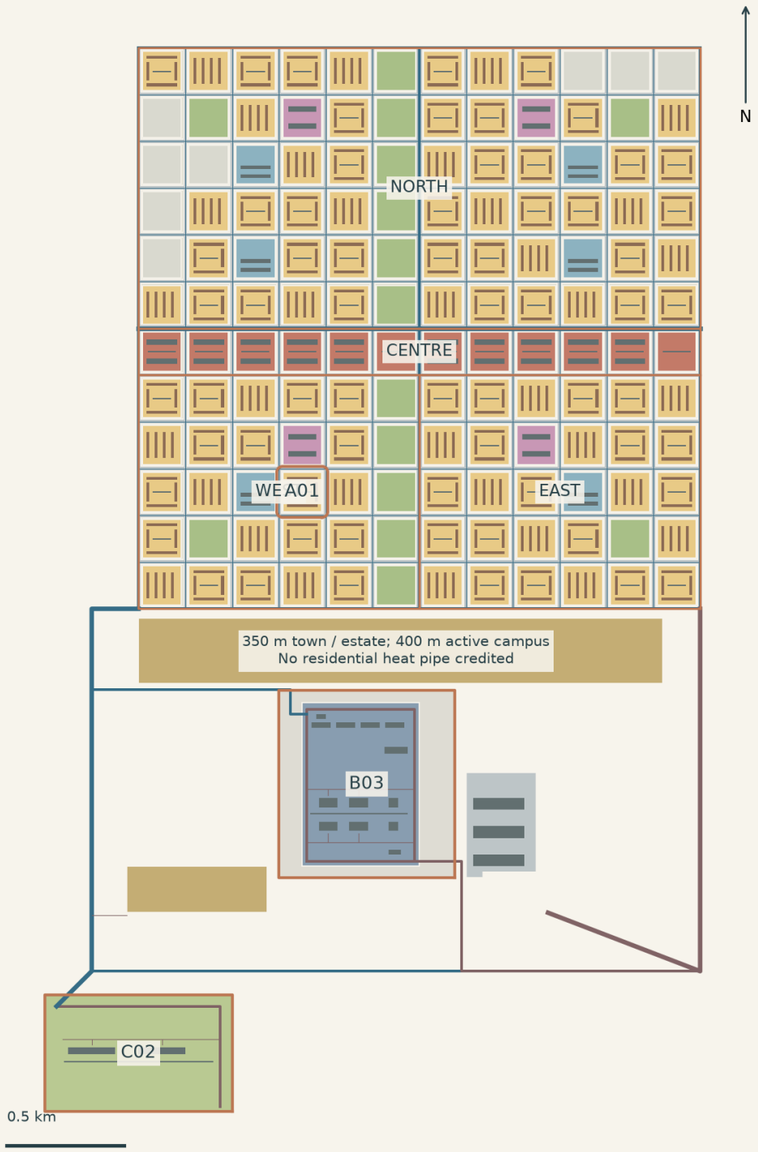


Detached bars and a 6 m deep shade canopy / conceptual level section

Both counts are declared design capacities, not sale plots, guaranteed household demand or surveyed building permissions. Landscape furniture and people in the finished images are illustrative.

Three places test three city promises

The nodes are located by the same coordinates as the master plan, not assembled into a fictional single view.



A01 / OPEN-COURT HOMES

West living district / P1. A complete residential court tests shared daily space, corner access and housing independent of an employer.

B03 / COMPUTE + REPAIR

Southern Works campus / P1-P3. Four staged modular groups test independent servicing and a limited local heat recipient. No claim that low-grade heat powers residential life.

C02 / FARM LEARNING EDGE

Southwest new trial campus / P1. Teaching, servicing and a public shade edge are arranged on new land outside the fixed pilots; the pilot interiors remain unchanged.

A01 / Open-court homes

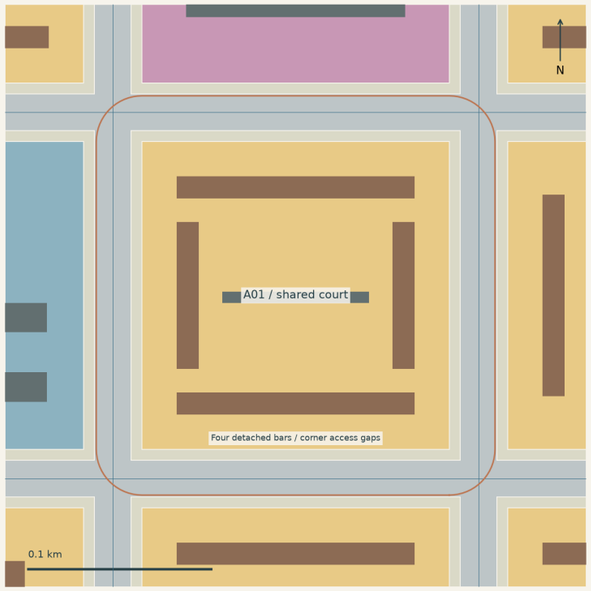
Three-storey housing, shared mineral court and an explicit shade band. Local aerial; full-build-out context.

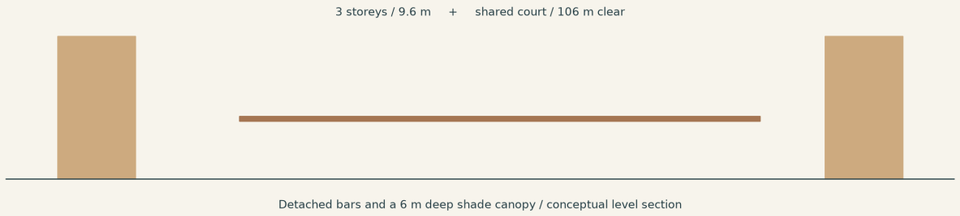


A01 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.







Detached bars and a 6 m deep shade canopy / conceptual level section

Residents enter through the open corners and meet under the existing shade structure. A housing cooperative or municipal operator must fund cleaning, accessible paving, lighting and canopy inspections. No cars are routed through the shared court.

B03 / An independent productive campus

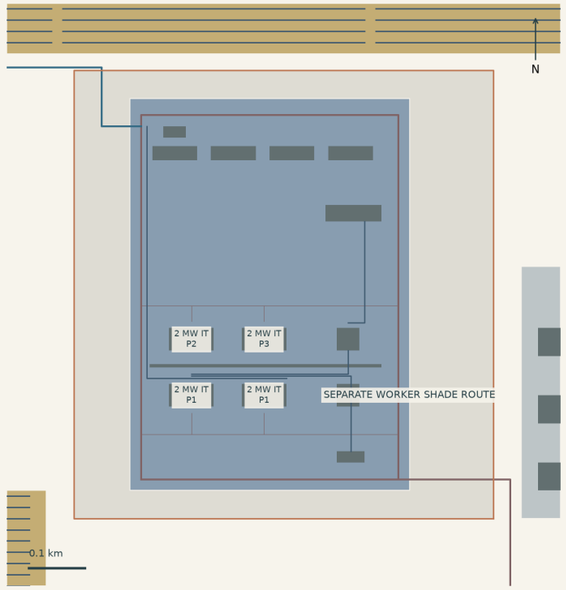
Four staged modular-service groups, repair sheds, utility servicing and a separate worker shade spine. Local aerial; all stages shown.



B03 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.





8 m service pavilions / 60 m service gap / 6 m worker shade

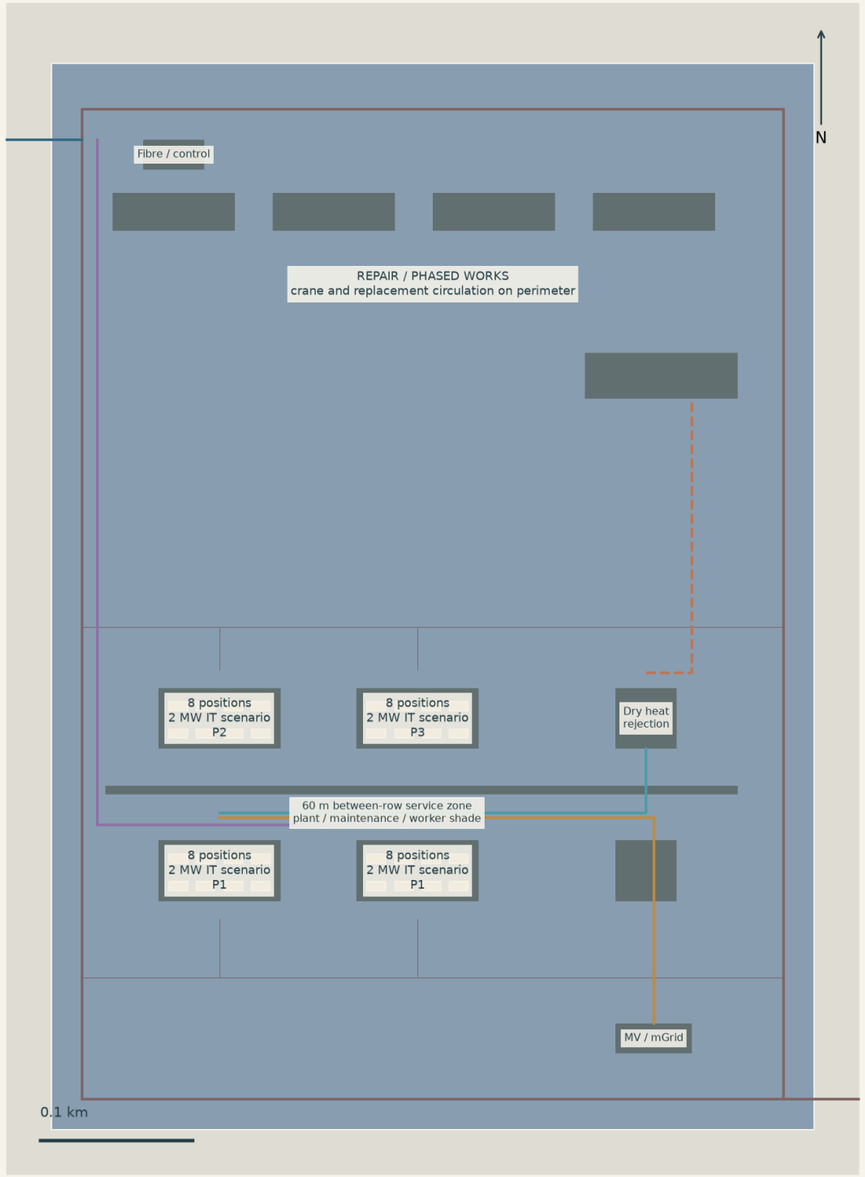


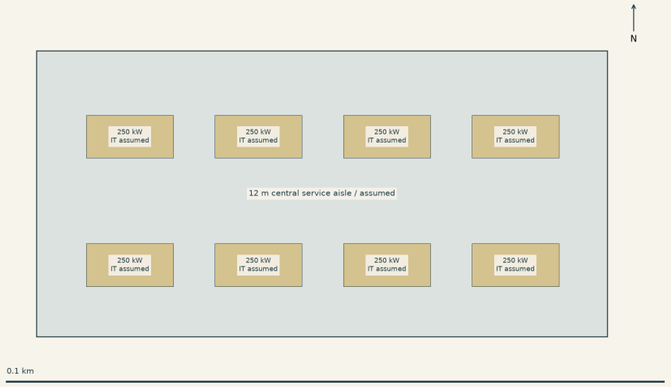
Covered modular positions; local heat test uncredited; no pipe to homes

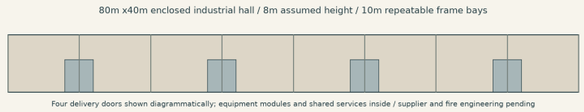
Workers reach the protected internal spine from the bus link. Freight takes the perimeter and service spurs. The registered human/machine interfaces are geometric flags, not proven-safe crossings. IT interruption contracts protect water/power limits; service budgets cannot depend on uninterrupted output.

B03 / compact modules and shared campus services

0.35 km2 serviced campus within0.60 km2 gross estate;0.25 km2 is unserviced future reserve with no capacity credit.





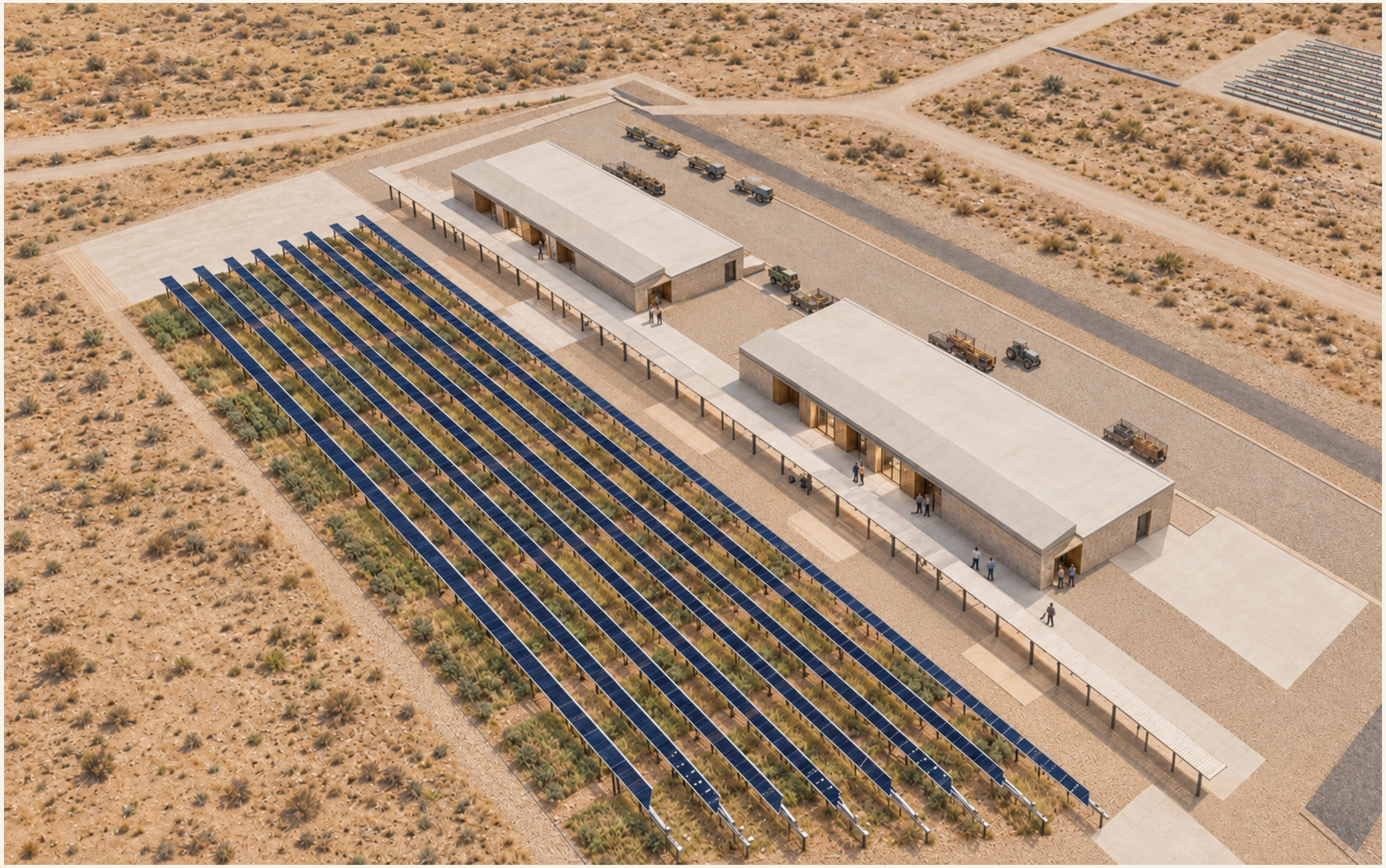


Four 80 x40m industrial halls each contain eight conceptual integrated equipment bays. Operating IT is assumed 250kW/position: P1/P2/P3 = 4/6/8 MW. Public XPOD reference: 6 racks x 56 kW =336 kW simple rack nameplate;1 MW power-block rating is not 1 MW IT.

50 m side gaps and 60 m between-row service space reserve delivery, replacement, shared plant/fibre/thermal routes and worker shade. Vendor layout, fire access and crane envelopes remain engineering checks. Heat rejection is provided; local drying earns no useful-heat credit yet.

C02 / A working farm-learning edge

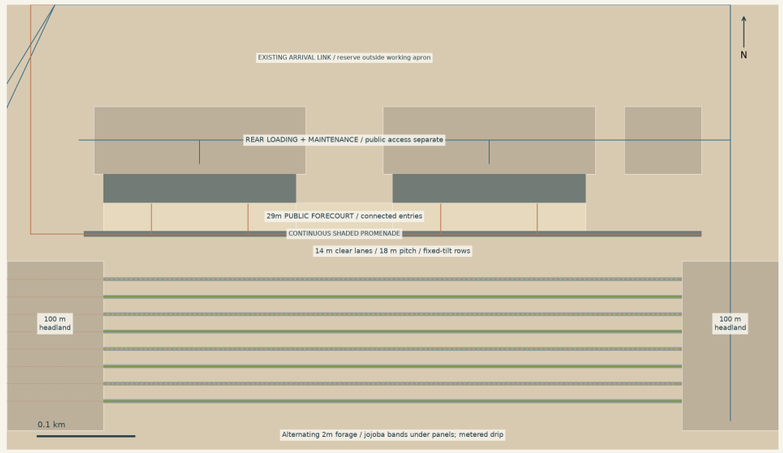
Two farm halls now meet a29m public forecourt and shaded promenade; rear loading remains separate. Close partial aerial; field ends cropped. Proposed forage/jojoba under8 rows; irrigation entitlement unconfirmed.

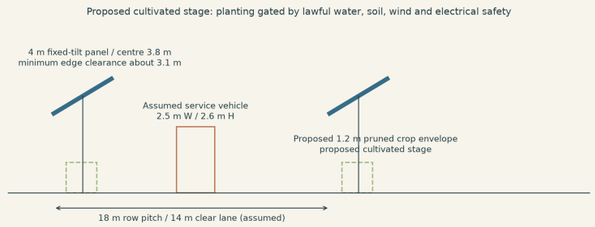


C02 / use, access and spatial detail

A separate geometry-derived eye-level camera, plan and conceptual section; image details are illustrative, not capacity evidence.



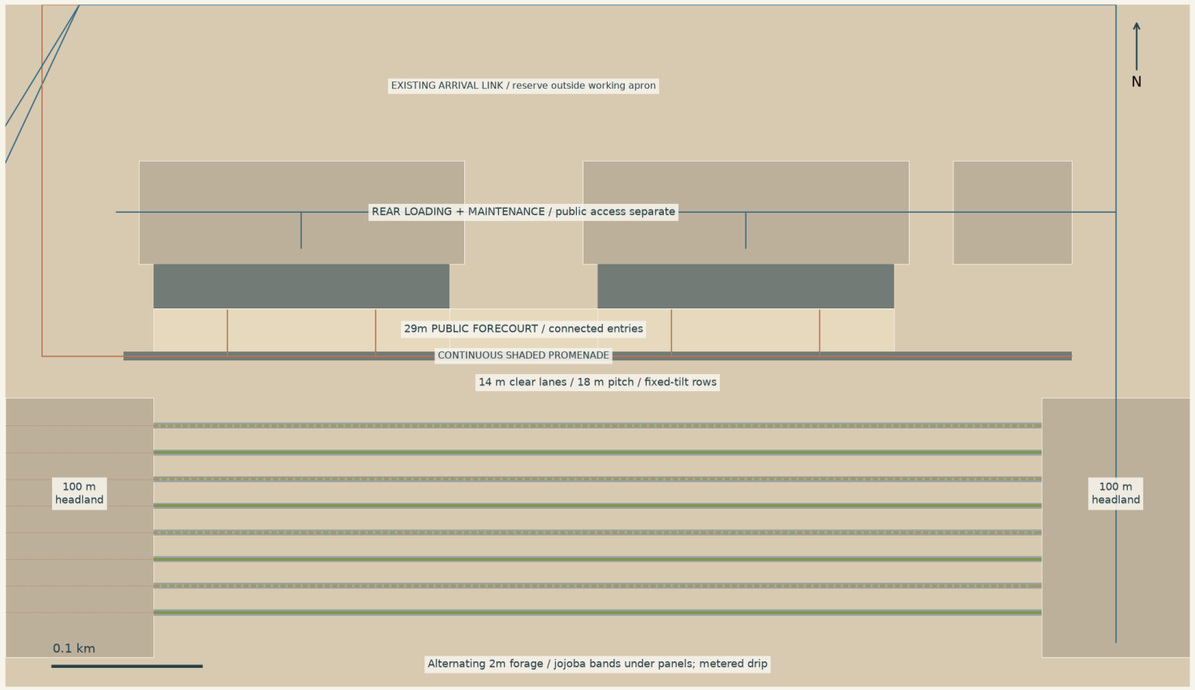




Visitors enter from the separated west approach, move under the canopy and cross the29m forecourts to workshop doors. Farm deliveries use the rear70m work aprons and northern service spur. Forage and pruned jojoba occupy alternating declared bands below raised panels. Planting requires lawful water and soil/wind trials; visitors stay outside the working lanes.

C02 / agriculture under the photovoltaic rows

Production, maintenance and visitors have distinct ground. This is proposed cultivation, not an existing planted field or a redesign of the protected pilots.



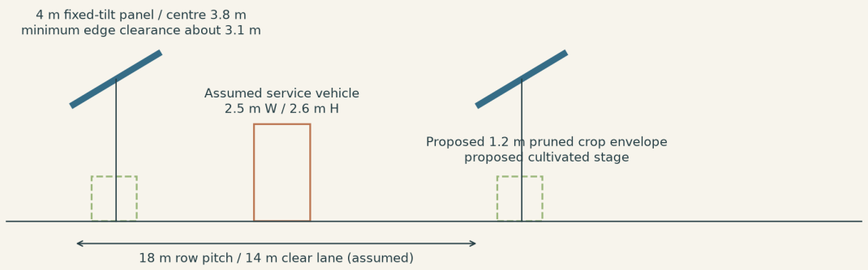
A connected working edge.
Two workshops move55m toward the public canopy. A29m front court serves entries and learning;70m rear aprons serve loading and maintenance. The crop field and100m turning headlands stay unchanged.

Two compatible crop systems.
Alternating 2 m bands carry forage and pruned jojoba. Four-metre panels have 3.8 m centre height and about 3.1 m minimum edge clearance; jojoba is held at 1.2 m. C02 has 18 m pitch, 14 m clear lanes and 100 m end headlands. All dimensions are assumed.

Wind and water.
Jojoba is wind-pollinated: provisional 1:8 male/female trial distribution along rows and both ends requires measured seasonal wind. Metered drip has separate valves and electrical isolation.

Real production budget.
New C02 plus main-field crop bands total 3.828 ha forage + 3.828 ha jojoba. Irrigation allowances 4000 /2000m3/ha/year give 22,968 m3/year. These are screening assumptions; lawful supply, soil and crop trials gate planting. No commercial yield credited.

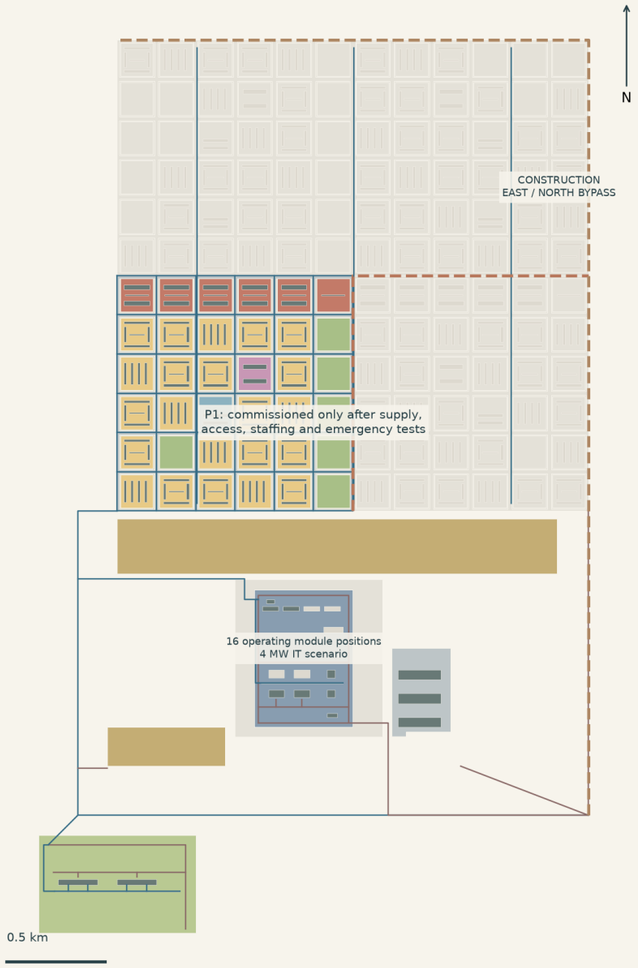
Proposed cultivated stage: planting gated by lawful water, soil, wind and electrical safety



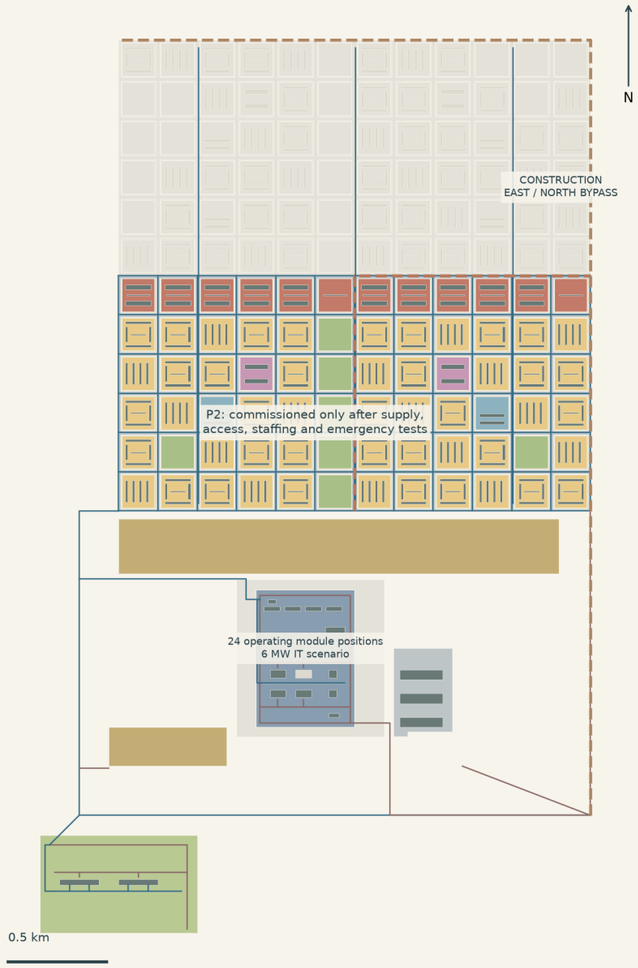
Commission operation, isolate the next works

Rust dashed routes reserve eastern/northern construction access; grey buildings remain unopened. These maps share the phase geometry and do not certify delivery rights.

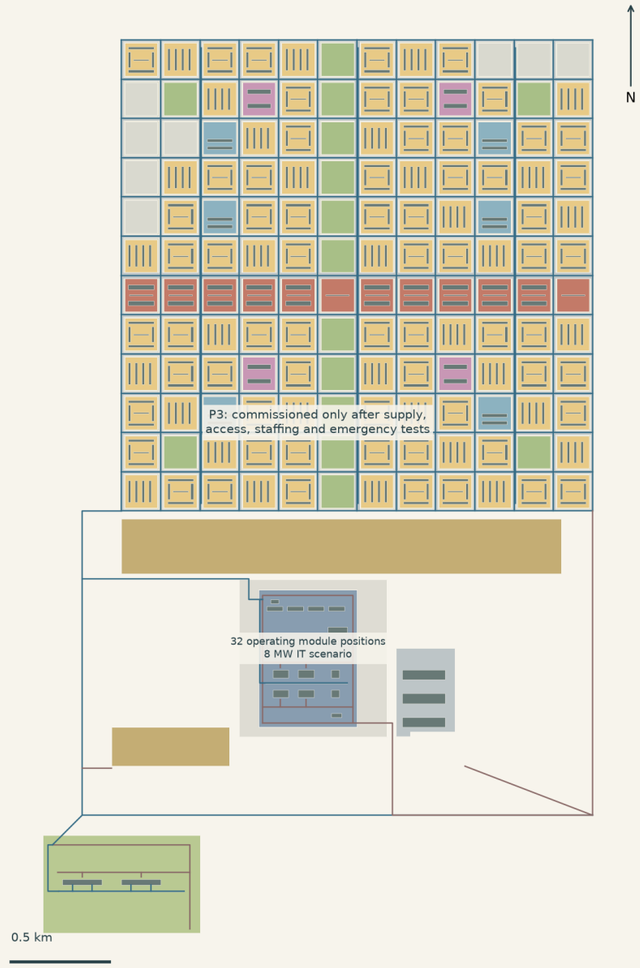
PHASE 1



PHASE 2



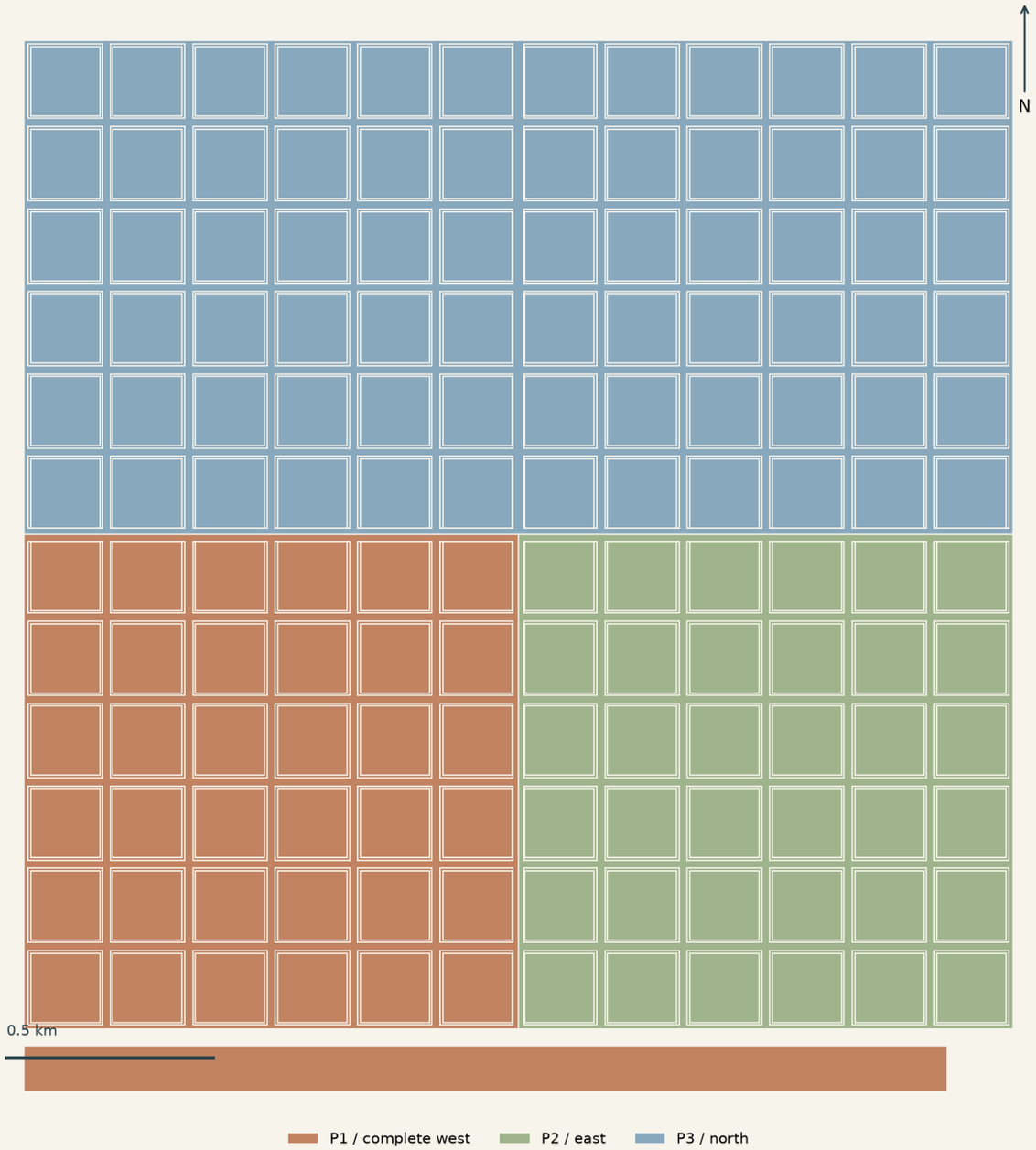
PHASE 3



Campus commissioning:16 /24 /32 positions at assumed 250kW operating IT each. Module delivery and essential-service testing precede energisation. Occupied West retains public access while future East/North works are fenced and bypassed.

Build a complete first phase; expand only when it works

Relative periods are scenarios. Phasing boundaries, homes, jobs and service obligations reconcile to the same building inventory.



P1 / Years 0-5 scenario

4,440 residents / 1,776 homes

Complete western neighbourhood, civic core, utilities and first productive halls. Funding and resource gates precede occupancy.

P2 / Years 5-10 conditional

9,720 residents / 3,888 homes

Eastern living district and additional productive capacity. Funding and resource gates precede occupancy.

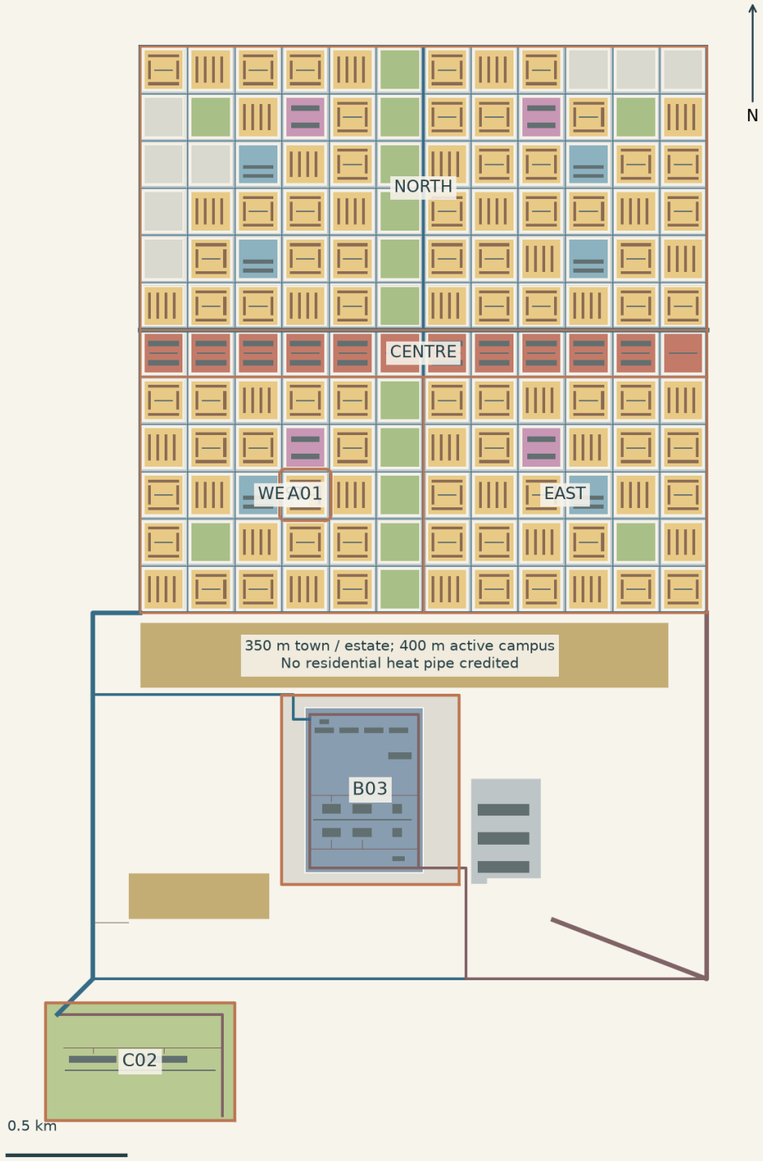
P3 / Years 10-20 conditional

19,680 residents / 7,872 homes

Northern neighbourhoods and final staged halls. Funding and resource gates precede occupancy.

The first neighbourhood must operate on its own

P1 scenario: 1,776 dwellings, 4,440 residents, 905 modelled jobs. These are separate capacities; no employment-demand balance is assumed.



Before the first keys

Lawful external access, water abstraction/supply, treatment, power and emergency access must be secured. The western school (750 places), care/shops and the western civic band open with occupied homes.

An operating budget, not a promise

Ring-fenced housing/public-service funding covers education, healthcare, street/shade maintenance and worker travel. Compute downtime cannot remove essential city services.

If later investment stops

The western neighbourhood retains its own school, daily care, markets, utility modules and access. Eastern/northern works are gated. The mismatch between 19,680 eventual residents and 2,150 modelled jobs remains a demand/employment verification need, not an assumed commuter solution.

Resource gates define what can be occupied

All figures below are declared scenarios derived from capacity; no water entitlement, grid contract, yield or climate measurement is supplied by these calculations.

WATER / FULL BUILD-OUT

774.0 thousand m3/y gross use

85 L/person/day domestic + 15 L/person/day civic, 35 L/job per workday +1,000m3/year panel cleaning, and an indicative 0.1 L/kWh compute makeup assumption. Reuse: 100,000 m3/y scenario. After 8% distribution loss, required fresh abstraction is 732,580 m3/y.

New forage/jojoba bands total7.656ha and include22,968m3/y assumed irrigation. Fixed pilot demand remains additional and unknown. No landscape irrigation or commercial crop yield is credited.

POWER / FULL BUILD-OUT

8 MW IT / four staged modular groups

P1/P2/P3 installed IT scenario: 4 / 6 / 8 MW. At assumed PUE 1.3, continuous full IT would use 91.1 GWh/y before town loads.

Explicit main PV rows reserve 26.784 MWp at an assumed 4 m panel width and 0.2 kWp/m2. At a synthetic 1,800 kWh/kWp/y yield, this is 48.2 GWh/y; it does not prove hourly matching or firm power.

Grid capacity, storage, weather and measured heat demand require independent study. No clean-energy percentage is claimed.

What this city chooses, and what it must still prove

The proposal trades shortest productive routes and maximum uninterrupted IT output for independent housing, public services and a clearer operating boundary.

A public-life decision

Living districts and the public band come first. Compute and production are arranged where service access can be controlled. This creates additional travel, utility cost and maintenance instead of claiming every adjacency is beneficial.

A capacity ceiling, not an instant new city

7,872 dwellings / 19,680 residents is the complete design capacity. Actual occupation requires housing demand, employment/livelihood evidence and independently funded services; the current 2,150 modelled jobs do not settle that question.

The next review has concrete questions

Survey and civil ground fit; lawful access and resource rights; noise, dust and emergency separation; an accessible walking/bus audit; crossing and canopy engineering; metered local drying demand; school staffing and annual operating budgets.

Traceable without overwhelming the presentation

Original controls, cameras, prompts, area/phase accounts and sources are in the separate technical appendix and machine files. Earlier V1 geometry is withdrawn from recommendation and retained for comparison. Neither polished images nor automatic file checks are independent design approval.